Customer Segmentation with K-Means: Applying the KDD Process

Introduction

Customer segmentation can help organizations explore differences in customer behavior and develop more targeted business strategies. Clustering is one way to discover groups in data when the records do not already have category labels.

In this project, I applied the Knowledge Discovery in Databases (KDD) process to the Mall Customer Segmentation dataset. I used K-means clustering to group customers according to annual income and spending score, then evaluated candidate cluster counts and interpreted the resulting groups.

Dataset & Research Question

Research Question: Can we group customers into useful clusters using their income and spending score?

The dataset was obtained from the Mall Customer Segmentation Data page on Kaggle. It contained 200 records and five columns:

- CustomerID
- Gender
- Age
- Annual Income (k$)
- Spending Score (1–100)

The spending score is a score provided in the dataset to represent customer spending behavior. The project focused on annual income and spending score. CustomerID was excluded from clustering because it is an identifier, not a behavioral measurement.

Following the KDD Process

1. **Selection**
   The dataset was selected because it contains income and spending information suitable for exploratory segmentation. Annual income and spending score were chosen as the two clustering features. Age and Gender remained in the original dataset for context but were not used to fit the model.

2. **Preprocessing**
   I checked for missing values, duplicate rows, duplicate customer IDs, and data types. The checks found no missing values in any column, no duplicate rows, and no duplicate customer IDs. All 200 records were retained.

3. **Transformation**
   The two selected features were standardized using scikit-learn's StandardScaler. After transformation, both features had means approximately equal to zero and standard deviations equal to one.

   Standardization helped put annual income and spending score on comparable scales before applying K-means.

4. **Data Mining**
   I tested K-means models with cluster counts from two through ten. The elbow method was used to compare inertia, while silhouette scores were used to assess how well customers fit within their assigned clusters relative to other clusters.

   The highest silhouette score among the tested values occurred at **five clusters**, with a score of **0.555**. The elbow plot also showed a substantial reduction in inertia through five clusters, followed by more gradual improvements. I therefore selected five clusters for the final model, using random_state=42 and n_init=10.

Interpretation & Evaluation

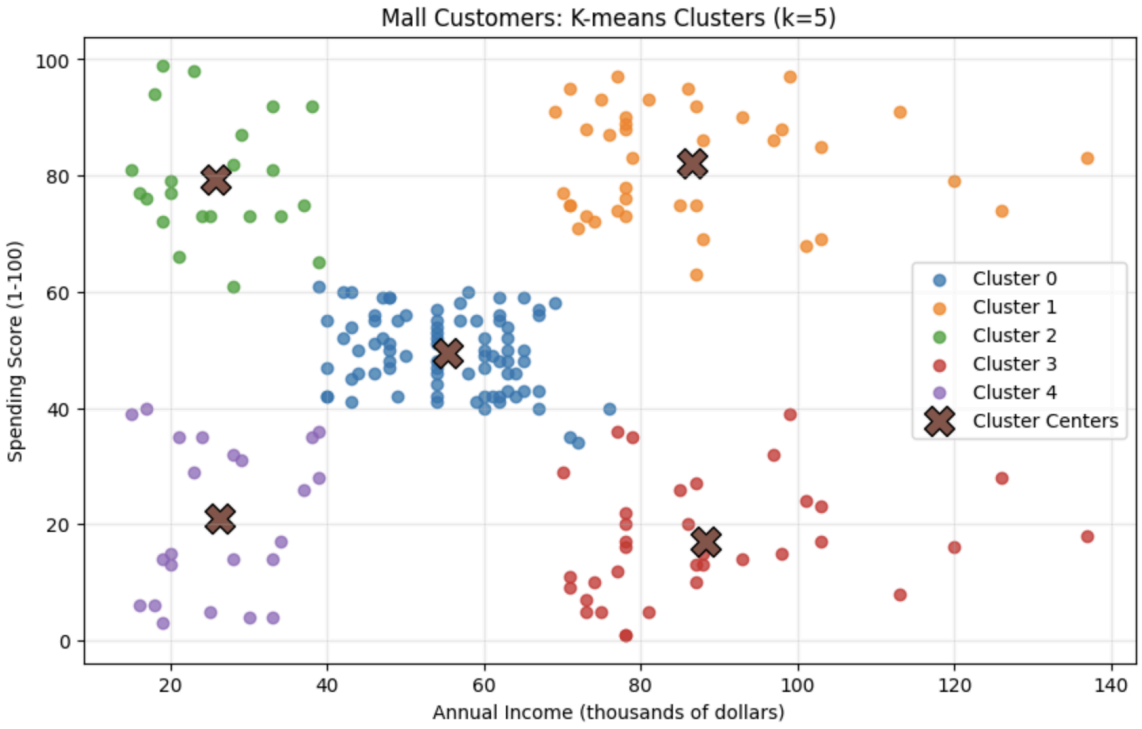


Figure 1. K-means customer segmentation with five clusters. Each dot represents a customer positioned by annual income and spending score. The colors distinguish the five clusters, and the large X-shaped markers indicate the cluster centers. The plot shows groups with different income and spending patterns; these are exploratory segments, not pre-existing labels in the dataset.

The final model assigned all 200 customers to one of five groups.

Cluster	# of Customers	Average Annual Income (k$)	Average Spending Score
0	81	55.30	49.52
1	39	86.54	82.13
2	22	25.73	79.36

3	35	88.20	17.11
4	23	26.30	20.91

The cluster visualization showed five groups:

- **Cluster 0 — Average income, average spending:** The largest group, with 81 customers.
- **Cluster 1 — High income, high spending:** 39 customers, potentially relevant for premium offers or loyalty programs.
- **Cluster 2 — Low income, high spending:** 22 customers, potentially responsive to affordable promotions.
- **Cluster 3 — High income, low spending:** 35 customers, a group for which further research could investigate low spending scores.
- **Cluster 4 — Low income, low spending:** 23 customers, potentially suited to budget-conscious offers.

These cluster labels are interpretations of the average characteristics of each group, not categories provided by the original dataset. The five-cluster model achieved a silhouette score of **0.555**, the highest among the tested values of two through ten clusters. This suggests reasonably well-separated groups for the selected features, although it does not establish that five clusters are the only valid solution.

Limitations

The analysis used only annual income and spending score. It did not use age or gender to form clusters, and the resulting groups do not explain the reasons behind customer spending behavior. K-means is also affected by feature selection, scaling, initialization, and the number of clusters chosen.

The results should be treated as exploratory rather than as proof that a particular marketing strategy will work. A business would need to validate the segments with additional behavioral data and real campaign outcomes before making major decisions.

Conclusion

The KDD process provided a step-by-step structure for selecting, preprocessing, transforming, mining, and evaluating the customer dataset. The final five-cluster model achieved a silhouette score of 0.555 and identified groups with different income and spending patterns.

This project demonstrates how unsupervised learning can support exploratory customer segmentation. The resulting profiles offer hypotheses for further business analysis, but additional validation would be needed before using them in real-world marketing decisions.

Project Artifacts

The notebook, detailed report, and clustered CSV are included in the KDD subdirectory of my repository: https://github.com/Olivia883/Cmpe255-data-mining-process/tree/main.